10/06/2025 BE IT SEM-VII C-SCHEME IRS QP CODE: 10082376

Time: 3 hours **Max. Marks: 80**

N.B

1. **Q.1 is compulsory**
2. **Attempt any three from the remaining five questions.**
3. **Assume suitable data, if required and state it clearly.**

Q 1 Attempt all. (20 marks)

a. Recall objectives of Information Retrieval Process
b. Differentiate between Information versus Data Retrieval.
c. Identify the various task performed by web search engine.
d. Importance of metadata and its types.

Q 2. Attempt all. (20 marks)

a. Show taxonomy of IR models and recall browsing models in detail.
b. Consider a very small collection C that consists in the following three documents:
   - d1: "beautiful garden"
   - d2: "evening garden time"
   - d3: "garden time is beautiful"

   Given the following query: "garden time", calculate the rank of each document using vector space retrieval model. (Use tf-idf vector for the query, and compute the score of each document in C relative to this query, using the cosine similarity measure.)

Q3. Attempt all. (20 marks)

a. Explain Boolean Models in detail with example.
b. Explain Natural language modelling issues and its solution

Q.4. Attempt all. (20 marks)

a. What is the purpose of using keyword based query? Briefly explain any 3 types of keyword based queries.
b. Apply Boyer Moore algorithm to construct bad match table and find the index of the given pattern for the string below with steps
   String: STUDENTS ARE SMART I LIKE SMART STUDENTS
   Pattern: SMART

Q 5. Attempt all. (20 marks)

a. Summarize various visualization techniques with respect to user interface design.
b. Construct a Suffixs, suffix trie and suffix tree, suffix array and supra index for the above sample text.
   Roses are red. Red roses are beautiful. Many people like red roses.

Q 6. Write short note on (20 marks)

a. Different types of Information Systems
b. Different markup languages and its applications.

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